Linux Commands

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  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

```
hp@Hima:~$ nano partical1.c
hp@Hima:~$ gcc partical.c -o partical
cc1: fatal error: partical.c: No such file or directory
compilation terminated.
hp@Hima:~$ gcc partical1.c -o partical1
hp@Hima:~$ ./partical1
Enter Linux command: uname
Parent PID : 785
Child PID : 789
Linux
Child process completed.
hp@Hima:~$ ./partical1
Enter Linux command: lscpu
Parent PID : 799
Child PID : 800
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          39 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   16
  On-line CPU(s) list:    0-15
Vendor ID:                GenuineIntel
  Model name:             12th Gen Intel(R) Core(TM) i5-1240P
    CPU family:           6
    Model:                154
    Thread(s) per core:   2
    Core(s) per socket:   8
    Socket(s):            1
    Stepping:             3
    BogoMIPS:             4223.99
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdts
                          cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
                          sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
                          bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
                          opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnmi umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
                          clear serialize ibt flush_l1d arch_capabilities
Virtualization features:
  Virtualization:         VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    384 KiB (8 instances)
  L1i:                    256 KiB (8 instances)
  L2:                     10 MiB (8 instances)
```

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```
Caches (sum of all):
  L1d:                    384 KiB (8 instances)
  L1i:                    256 KiB (8 instances)
  L2:                     10 MiB (8 instances)
  L3:                     12 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-15
Vulnerabilities:
  Gather data sampling:       Not affected
  Ghostwrite:                 Not affected
  Indirect target selection:  Not affected
  Itlb multihit:              Not affected
  L1tf:                       Not affected
  Mds:                        Not affected
  Meltdown:                   Not affected
  Mmio stale data:            Not affected
  Old microcode:              Not affected
  Reg file data sampling:     Mitigation; Clear Register File
  Retbleed:                   Mitigation; Enhanced IBRS
  Spec rstack overflow:       Not affected
  Spec store bypass:          Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:                 Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:                 Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds:                      Not affected
  Tsa:                        Not affected
  Tsx async abort:            Not affected
  Vmscape:                    Not affected
Child process completed.
hp@Hima:~$ ./partical1
Enter Linux command: lsblk
Parent PID : 812
Child PID : 813
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                                /
Child process completed.
hp@Hima:~$ ./partical1
Enter Linux command: ps
Parent PID : 823
Child PID : 824
```

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hp@Hima: ~

```
Child PID : 824
    PID TTY          TIME CMD
    649 pts/2    00:00:00 bash
    823 pts/2    00:00:00 partical1
    824 pts/2    00:00:00 ps
Child process completed.
hp@Hima:~$ ./partical1
Enter Linux command: top
Parent PID : 834
Child PID : 835
top - 08:27:41 up 15 min,  1 user,  load average: 0.05, 0.03, 0.00
Tasks:  29 total,   1 running,  28 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7775.7 total,   7088.0 free,    606.8 used,    232.8 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7168.9 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24204  15348  11548 S   0.0   0.2   0:00.73 systemd
      2 root      20   0    3180   2204   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
      7 root      20   0    3180   2048   1940 S   0.0   0.0   0:00.00 init
     47 root      19  -1   50420  16640  15380 S   0.0   0.2   0:00.25 systemd-journal
     79 systemd+  20   0   22416  14524  12032 S   0.0   0.2   0:00.10 systemd-resolve
     85 root      20   0   35424  12336   9192 S   0.0   0.2   0:00.49 systemd-udevd
    108 root      20   0    2888   1944   1820 S   0.0   0.0   0:00.06 chronyd-starter
    112 root      20   0    4472   3092   2844 S   0.0   0.0   0:00.01 cron
    118 message+  20   0    8792   5384   4708 S   0.0   0.1   0:00.06 dbus-daemon
    161 root      20   0   44096  29796  14652 S   0.0   0.4   0:00.31 networkd-dispat
    172 root      20   0   18436   9428   8184 S   0.0   0.1   0:00.13 systemd-logind
    194 syslog    20   0  220548   5720   4440 S   0.0   0.1   0:00.08 rsyslogd
    238 _chrony   20   0   23868  11192   7068 S   0.0   0.1   0:00.08 chronyd
    247 _chrony   20   0   12096   2400   1760 S   0.0   0.0   0:00.00 chronyd
    285 root      20   0  123460  32664  18056 S   0.0   0.4   0:00.20 unattended-upgr
    291 root      20   0    5492   2724   2536 S   0.0   0.0   0:00.01 agetty
    342 root      20   0    3184   1108    980 S   0.0   0.0   0:00.00 SessionLeader
    343 root      20   0    3200   1248   1108 S   0.0   0.0   0:00.05 Relay(344)
    344 hp        20   0    6268   5520   3848 S   0.0   0.1   0:00.04 bash
    345 root      20   0    8644   5544   4704 S   0.0   0.1   0:00.00 login
    388 hp        20   0   22340  13548  10932 S   0.0   0.2   0:00.11 systemd
    390 hp        20   0   22912   4088   2088 S   0.0   0.1   0:00.00 (sd-pam)
    420 hp        20   0    6136   5420   3864 S   0.0   0.1   0:00.02 bash
    543 hp        20   0    8156   6060   3920 S   0.0   0.1   0:00.57 top
    644 root      20   0    3184   1116    980 S   0.0   0.0   0:00.00 SessionLeader
    645 root      20   0    3200   1256   1108 S   0.0   0.0   0:00.13 Relay(649)
    649 hp        20   0    6268   5600   3928 S   0.0   0.1   0:00.07 bash
```

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Copying the File :

```
hp@Hima: ~
GNU nano 8.7.1                                                              pa
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
int src, dest;
char buffer[1024];
int bytes;

src = open("file1.txt", O_RDONLY);

if(src < 0)
{
printf("Cannot open source file\n");
return 1;
}
dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

while((bytes = read(src, buffer, sizeof(buffer))) > 0)
{
write(dest, buffer, bytes);
}
close(src);
close(dest);

printf("File copied successfully.\n");

return 0;
}
```

```
   20 | while((bytes = read(src, buffer, sizeo
      | ~~~~~~
hp@Hima:~$ nano partical1.c
hp@Hima:~$ gcc partical1.c -o partical1
hp@Hima:~$ ./partical1
File copied successfully.
hp@Hima:~$
```